

CaneMap System - User Manual (Updated)

CaneMap: A Location-Based Sugarcane Growth Tracker with Harvest Prediction Features

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1. Introduction and System Fundamentals

1.1 What is CaneMap?

CaneMap is a centralized and role-based sugarcane management system designed to assist Field Handlers, Caretakers, Sugar Regulatory Administration (SRA) Officers, and System Administrators in monitoring and managing sugarcane field operations through digital technologies.

The system provides real-time monitoring of field activities, GIS-based mapping, growth tracking, harvest prediction, reporting, analytics, and notifications using both web and Android platforms.

1.2 Key Features

- **GIS-Based Field Mapping and Boundary Visualization** – Allows users to map sugarcane fields digitally and visualize exact field boundaries using GIS technology.
 - **Role-Based Access Control (RBAC)** – Restricts system access based on user roles to ensure data security and proper authorization.
 - **Field Registration and Approval Workflow** – Enables field handlers to register fields while allowing administrators or SRA officers to review and approve submissions.
 - **Growth Tracking Timeline** – Provides a timeline view of sugarcane growth stages and recorded field activities.
 - **Harvest Prediction using DAP (Days After Planting)** – Estimates the expected harvest period based on the number of days after planting.
 - **Task and Activity Monitoring** – Helps monitor assigned tasks, field activities, and their completion status.
 - **Weather Forecast and Advisory** – Displays weather updates and advisories to assist users in planning field operations.
 - **Notifications and Alerts** – Sends reminders and alerts regarding important activities, updates, or deadlines within the system.
 - **Report Generation and Export** – Allows users to generate and export reports for documentation and analysis purposes.
 - **Analytics Dashboard** – Presents summarized data and visual insights to support monitoring and decision-making.
 - **Offline Record Logging and Synchronization** – Enables users to record field data offline and synchronize it once internet access is available.
 - **Real-Time Monitoring and Cloud Synchronization** – Supports real-time data updates and cloud synchronization for accurate and accessible field information.
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1.3 System Requirements

- Modern Web Browser (Chrome, Firefox, Edge, Safari)
- Android Mobile Device (Android 10–12 Recommended)
- Internet Connection (recommended for synchronization and notifications)
- Registered User Account

1.4 User Roles and Access Levels

Role Name	Role Code	Access Level	Primary Dashboard	Key Capability
System Administrator	admin	Full System Administrator	Admin Dashboard	Manage users, roles, system operations, and cloud services
SRA Officer	sra_officer	Monitoring and Validation	SRA Dashboard	Approve field registrations, review reports, monitor operations
Field Handler / Caretaker	caretaker	Field Operations	Field Handler Dashboard	Register fields, monitor growth, encode activities, submit reports

2. Getting Started and Navigation

2.1 Accessing the System

1. Open the CaneMap website or Android application.
2. Wait for the homepage to load.
3. Click either:
 - Login
 - Register

2.2 Account Registration

1. Click the “Register” button.

2. Fill out the registration form completely:

- Full Name
- Email Address
- Password
- Confirm Password

3. Click “Create Account” or “Submit”.

Note: Use an active email address because email verification is required.

2.3 Email Verification

1. Open your email inbox.
2. Locate the verification email from CaneMap.
3. Open the email and click the verification link.
4. Wait for the confirmation message.

Important: Users must verify their email before accessing full system features.

2.4 Logging In

1. Enter your registered Email Address.
2. Enter your Password.
3. Click “Login”.

After successful login, the system redirects users to their role-specific dashboard.

2.5 Dashboard Navigation

After login, users are redirected based on their assigned role:

Role	Dashboard
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System Administrator	Admin Dashboard
SRA Officer	SRA Dashboard
Field Handler / Caretaker	Field Handler Dashboard

2.6 Sidebar Navigation

The left sidebar provides access to all modules:

- Dashboard
- Field Registration
- Map Visualization
- Growth Tracker
- Task Management
- Reports
- Analytics
- Notifications
- Weather Advisory
- Profile Settings

Features:

- Collapsible Sidebar
- Active Page Indicator
- Quick Navigation Links

3. Dashboard Overview (By Role)

3.1 Field Handler Dashboard

Key Sections:

1. Registered Fields Overview
2. Growth Tracking Timeline

3. Harvest Prediction Summary
 4. Task Monitoring
 5. Weather Advisory
 6. Notifications Panel
 7. Analytics Summary
 8. Recent Activities
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3.2 SRA Officer Dashboard

Key Sections:

1. Pending Field Applications
 2. Registration Approval Statistics
 3. Interactive GIS Monitoring Map
 4. Submitted Reports
 5. Harvest Monitoring
 6. Notifications and Requests
 7. Operational Analytics
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3.3 System Administrator Dashboard

Key Sections:

1. User Management Overview
2. System Performance Monitoring
3. Database and Cloud Services Status
4. User Activity Logs
5. Security Monitoring
6. System Notifications

4. Core Workflow: Field Registration and Mapping

4.1 Field Registration

Who Can Access: Field Handler / Caretaker

4.1.1 Registering a Field

1. Navigate to “Field Registration”.
2. Click “Add Field” or “Register Field”.
3. Fill out all required information:
 - Field Name
 - Field Owner Name
 - Corporation Name
 - Barangay / Location
 - Sugarcane Variety
 - Terrain Type
 - Field Size
 - Additional Details
4. Upload required documents:
 - Ownership Documents
 - Verification Files
 - Field Images
5. Use the map to:
 - Locate the field
 - Draw boundaries
 - Confirm field location
6. Click “Submit Registration”.

4.1.2 Monitoring Registration Status

Possible registration statuses include:

- Pending – Waiting for SRA review
- Approved – Registration accepted
- Review – Requires additional checking
- Rejected – Invalid or incomplete information

Notifications are automatically sent after status updates.

4.2 GIS Map Visualization

Who Can Access: All Roles

Viewing Registered Fields

1. Navigate to “Map Visualization”.
2. View:
 - Field Locations
 - Field Boundaries
 - Growth Status
 - Expected Harvest Dates
3. Click field markers to view detailed information.

The system uses Leaflet.js and OpenStreetMap for geographic visualization.

5. Core Workflow: Growth Tracking and Harvest Prediction

5.1 Growth Tracking Timeline

Who Can Access: Field Handler / Caretaker

Recording Growth Activities

1. Navigate to “Growth Tracker”.

2. Select a registered field.
3. Record growth activities such as:
 - Land Preparation
 - Planting
 - Replanting
 - Ratooning
 - Maturity
4. Save the activity log.

The timeline automatically organizes records chronologically.

5.2 Harvest Prediction

The system computes harvest schedules using:

- DAP (Days After Planting)
- Planting Date
- Ratooning Data
- Sugarcane Variety Information

Features:

- Estimated Harvest Dates
 - Harvest Readiness Monitoring
 - Harvest Notifications and Reminders
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5.3 Harvest Monitoring

1. Navigate to “Harvest Monitoring”.
2. Encode:
 - Harvest Date
 - Harvest Weight
 - Number of Trips

- Harvest Remarks

3. Save the harvest record.

The system stores records for analytics and reporting.

6. Core Workflow: Task and Activity Management

6.1 Task Management

Who Can Access: Field Handler / Caretaker

Operational Tasks Include:

- Land Preparation
 - Planting
 - Fertilization
 - Irrigation
 - Harvest Preparation
 - Post-Harvest Activities
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6.1.1 Encoding Task Records

1. Navigate to “Task Management”.
2. Select an operational task.
3. Enter operational details.
4. Set task status:
 - In Progress
 - Done
5. Save the record.

Important:

- Tasks follow a sequential workflow.
 - Completed tasks are automatically removed from available options.
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6.2 Date Validation

When encoding activities:

- Only current and future dates are allowed.
 - Past dates are restricted to maintain data consistency.
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6.3 Offline Logging and Synchronization

When internet connection is unavailable:

- Users may continue encoding activities offline.
- Data is temporarily stored locally.

Once internet is restored:

- The system automatically synchronizes data with the cloud database.
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7. Supporting Features (Reports, Notifications, Weather, Analytics)

7.1 Reports Module

Features:

- Create Reports
- Edit Reports
- Download Reports
- Export Reports in:
 - PDF
 - CSV

Steps:

1. Navigate to “Reports”.
 2. Click “Create Report”.
 3. Enter report details and attach supporting files.
 4. Save and submit the report.
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7.2 Analytics Dashboard

Displays:

- Growth Progress
- Harvest Status
- Activity Completion
- Field Statistics
- Harvest Weight
- Transport Trips

Analytics help improve operational planning and monitoring.

7.3 Weather Advisory

Displays:

- Current Weather
- Rain Forecast
- Temperature
- Field Work Advisories

The module helps users safely plan agricultural activities.

7.4 Notifications System

Types of Notifications:

1. Field Registration Updates
2. Task Progress Alerts
3. Growth Milestones
4. Harvest Reminders
5. Report Feedback

Using Notifications

1. Click the notification icon.
2. Select a notification to redirect to the related module.
3. Notifications are automatically marked as read after opening.

8. Administrative Functions

8.1 User Management

Who Can Access: System Administrator

Functions:

Task	Steps
Create User	Click “Add User”, enter user details, assign role, and save
Edit User	Select a user, update information, and save changes
Deactivate User	Select user and confirm deactivation
Reset Password	Reset user password for account recovery

8.2 Role Management

The administrator may:

- Assign User Roles
- Modify Access Permissions
- Monitor User Access Levels

8.3 System Operations Monitoring

The administrator may:

- Monitor system performance
 - Maintain cloud synchronization
 - Manage database records
 - Update system configurations
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9. Common Tasks and Troubleshooting

9.1 Changing Your Password

1. Open your Profile Menu.
 2. Select “Change Password”.
 3. Enter:
 - Current Password
 - New Password
 - Confirm Password
 4. Click “Save Changes”.
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9.2 Viewing Registration Status

1. Navigate to “Field Registration”.
 2. View all submitted registrations and their statuses.
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9.3 Viewing Reports and Activity Logs

1. Navigate to “Reports” or “Analytics”.
 2. View historical records and activity summaries.
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9.4 Common Issues and Solutions

Issue	Solution
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Cannot login	Verify email and password or reset password
Field registration rejected	Review remarks and upload missing requirements
Map not loading	Check internet connection and refresh the page
Notifications not appearing	Refresh the application or reconnect to the internet
Offline data not syncing	Reconnect to the internet and reopen the application

9.5 Getting Help

Contact Information:

- Office: Sugar Regulatory Administration (SRA) Office
- Support Hours: Monday–Friday, 8:00 AM – 5:00 PM

For technical concerns, provide:

- Registered Email
 - User Role
 - Detailed Description of the Issue
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10. Appendix (Reference)

Appendix A: Supported File Formats

Type	Supported Formats	Recommended Size

Document Files	.pdf, .doc, .docx, .txt	Under 10 MB
Spreadsheet Files	.xlsx, .xls, .csv	Under 10 MB
Image Files	.jpg, .jpeg, .png, .webp	Under 10 MB

Maximum recommended upload size: 50 MB per file.

Appendix B: System Policies

The system follows:

1. Role-Based Access Control (RBAC)
 2. Secure Authentication and Email Verification
 3. Structured Field Registration and Approval Workflow
 4. Data Privacy and Secure Cloud Synchronization
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Appendix C: Security Guidelines

Users should:

- Keep passwords confidential
- Avoid account sharing
- Verify field information before submission
- Logout after system use
- Maintain stable internet connection for synchronization